

What is your current role? What do you work as? Dev? Technical Lead

I think it's at a transition moment where my role is kind of strange, but I think it can be technical leadership, but still acting strongly in development and participating a lot in all the architectural, team decisions, etc.

And how long have you worked with software development?

Since the pandemic, I think 6 years.

And which AI tools do you use at work?

Nowadays we have been using Copilot, mainly for pull request correction, right, on GitHub. We also use Claude Code for code generation. Code has been, we have adopted an AI First posture in the company since the beginning of this year, so we always try to generate all the code with Claude. Eh, I have also used Gemini for the same function, also for code generation, at the time when we were still experimenting, like, which one would be official. These are the main ones.

And how often do you use these tools?

Every day.

And what types of activity do you usually use AI for?

Code generation, card description generation, PR review, as I said. I have also used it to discuss solutions for something, there was a problem there and then I said how I was thinking of doing it and asked if it agreed, if it had any other suggestion, etc. to discuss alternatives.

Can you describe a recent case in which you used AI to perform some development activity. Like, from the beginning until having the expected final result.

As I said, we have been using it 100% of the time like this, every day. So, in several recent cases. One that we can discuss now is trying to define a screen. And then it goes from understanding to generating the specifications of the screen, as well as beyond code, there are some things specifically for apps, in this case, such as, for example, instrumentation, for us to put there a sequence of events that describe the user's behavior in the application. This is usually already what we also analyze. Issues such as accessibility, there is a button, but will the reading from the device's accessibility helper be able to read this button? And what will it read as well? If it will help. So, when it is for front-end, we usually have a few extra things.

And then that's where we come in with, okay, I have to make this screen, I have the Figma there with its definition and then I put, when I have a token I put Copilot to use Figma's MCP, not Copilot, Claude Code to use Figma's MCP and read Figma and then generate the code for this screen and make a little document that is this card list with a lot of requirements. Actually, with several specifications even, accessibility issues, as I said, and everything else. And when it is some screen that is a little newer, let's say, not even that complex, but it can be said to be complex, such as, for example, it will call a route that does not exist yet. Then it is also nice to use AI to generate the tasks for the other cards. So, like, I have to make a screen that calls a route. This route does not exist yet, so I will make another specification to generate the code for this route and then, when I do it at the same time, I already guarantee, actually, the AI will already guarantee that it will work, because, like, the application will call the route with the right parameters, the right name. We will use distributed architecture for this route. So, this route is in the BFF, and it will call the microservice. It has to implement it there in the microservice too, there is the same contract issue. If it is a route under the external partner, there is still from the microservice to the partner. And so it goes. So, when it is something kind of chained, it is good to do everything at once, the specification from beginning to end to ensure that everything talks to each other, because otherwise, that thing, you make a screen and then when you look at the partner's documentation, you find out there is no way to do a filter that was on the screen. Then you have to go back redoing it. So it is nice to use AI to have this very macro view and that is how we have liked doing it more. It is, taking a task from

beginning to end and starting to break it down. I like breaking it down by repository, so a more technical specification for an app, another for the BFF, another for the service, and another for something else. But ensuring that everything is generated there connecting so that there are not so many fixes to do afterward. Even though it happens, right, some little thing or another that came out wrong.

So basically, just so I understand how your flow works. There is a prototype and then you pass it to AI to generate some requirements specification, is that it? And then from that specification you will generate your code, everything you need.

Usually it is like this, we take something to deliver. I even have a PM, right? She has even been making the definition with a PRD and such, but lately I have been using Figma more, which is what I use to discuss with people and understand the solution that will be made beforehand. And then after that I put the Figma link and the thing starts coming out, like that. Usually, the requirements and everything already, I myself ensure what happens.

When you use AI in a task, how do you define what needs to be done, like? Do you already have this clear beforehand or do you build it throughout the interesting things?

There are two things. The first is that in my company we have a team specialized in AI, which is to do things that support us so that we can become an AI First company more quickly and effectively. So, this team, they made a little library that has several markdowns, that is, basically several prompts. And, for example, there they made a markdown with the instructions and scripts to generate a card description, which we call PRD. So, there are already some things described with these specifications of what the PRD needs to have, or else a specification too, what it needs to have. And then, but we always add too, right? depending on what it is. At the time of generating this document, it will generate based on this template, these instructions, and then we add things. But the idea is that this template should be something living, that like, we realize that, wow, every time I have to tell it to remember accessibility on the screen. So, we are encouraged, by the company itself, to contribute and go directly to this repository and put: "Do not forget to put accessibility so that all screens have it." So, it is a certain middle ground, but mainly at this moment of building this template, we have had a lot of both, but I think the idea is that we have the minimum intervention from now on, this is decreasing over time, as we learn to make these prompts more complete, right? So we don't have to talk about accessibility every time, talk about some screen alignment every time, which it always gets wrong and then, as I said, we are encouraged to do this so everyone can use it and grow quickly in the use of AI.

So, basically, you have a problem there, you pass the description and it already sets everything up for you.

Yes, yes, because there is already a ready-made prompt, right? So, it is just a way to make it a little easier than my having to copy it every time, but they made a CLI that when I open my VS Code of life, I can press slash, then there is a bunch of comments that was what we made, based on this CLI. So, then there is slash, AI PRD. Then it will load the prompt with the instructions to create a PRD. After this slash, I already, actually, I do slash and AI PRD, write everything, then when I send it, Claude will already include there in its context and this PRD template. And then another cool thing is that there is an instruction that you give to Claude for it to use a kind of dynamic VS Code window and then it keeps asking us questions. This is also very well specified, if there is any open question, ask me. And then questions keep appearing with options, which is very multiple choice, with an "other" field and then it becomes a real interaction, which is even easier to visualize.

And has the use of AI changed the way you work, if so, how?

It changed a lot, right? Because there are even a lot of, some videos with memes about this, about how when we worked before it was full, typing and thinking about the solution and now it is telling

Claude to do it, while it does it we are staring at nothing, like, watching time pass or finding something else to do. I think this is also a consequence to think about, because before I was there with a task, and I spent a day focused on that task. So, I was with it. Nowadays, I give a focus time to that task, which is mainly when generating all these documentations, these specifications. While Claude is generating code, I get this feeling that I need to take another task, because I can be more productive. So, then I take a second one, and while the second is generating, I look at the first and it is still generating, so, I keep trying to take a third. And so, it gets to a point where I start to get tangled up, you know? Like, I have about three agents open here, each of them is waiting for me to answer something. And then, I felt, right at the beginning when we started using AI, it was much more tiring for me, I ended the day literally more tired than before when I had to spend the day thinking. Because I think it generates this expectation of such high productivity, then we start trying to keep several plates spinning at the same time. And even though the AI is generating automatically, there is still a cognitive load, you know? Of being able to keep up with all of this, because now you even feel bad about spending half the day looking at the screen doing something and you do nothing. So I think at least for me it creates a temptation to try to do something else too, to give agility and increase task throughput, etc. But then there is a balance. I have already understood now, no, even if I am not doing anything this minute, I will not take another task, otherwise I will go crazy. With how many things I am already doing at the same time. So, like, it changed a lot, not only in the way the work is actually done, but my professional life was impacted, you know? I got more tired over time. I had to reflect on everything that I am, everything you are going to do, you have to stop and ask yourself if you are not doing something that an AI should do. Any process starts to have this voice in your head that I should be making a prompt for this, I should be making an agent to do this and not do it myself, so that I don't do it and it does it for me. And then it is a very big change in the way we think too, you know?

Today, what do you feel you are dedicating effort more to?

I think there are two big things. The first is this moment of specification. So, as I said, we have this first stage of writing what needs to be done, then after that writing how it needs to be done, then there it already goes into a more code-oriented part so it can be generated in a more, uh, assertive way. But then, a good part of the time is writing these PRDs and reviewing, so, reviewing a lot of text files. Instead of reviewing PR, now I have to review the plan that will generate PR. And then there is this first stage, but lately I have also been doing a lot of correction. So, like, Claude generated a bunch of things there, I took a look like this, it seemed all right to me, then a mini correction has to be submitted. Then it will, I will have to open a new PR, a new prompt, wait again, a mini correction. Then another mini correction. So this is even something that we have, were discussing this week, about how, okay, AI generates code quickly. But doesn't this code need to be improved a lot afterward? Is only generating code quickly an interesting metric or else only the number of open PRs or number of commits made because AI increased the number of PRs closed in the team. This does not seem totally reliable, you know? Then we generate a lot, for example, one hypothesis is that we generate a lot of PR because we have to fix a lot of things that the AI generated. And that is why we generate more than we generated before. So, uh, it definitely changed the focus that before was writing, nowadays it is purely reviewing. Whether reviewing a document with a plan or with the specifications of something or reviewing the code that it generated, which also has to happen. So, it is much more a life of a lot of reading and review and a screen split into several parts to keep up with everything it is doing, than how it was before, which was typing.

Comparing before and after, what changed the most for you?

There are several things. Nowadays, mainly, we try to think at the process level. I think it has already become very clear that it is like this. Actually, this became an obligation, to tell you the truth, it is an instruction that my company gave from a certain point, which is, all code has to be generated with AI. So, okay. That part is easy, we have to stop typing code and start writing

instructions. Up to this point I really think it is easy to start doing. Ah, okay, there is skill, there is how to make a prompt, I don't know what, but these are evolutions to improve something that is easy to do. Now, what I think is difficult, which has been a bigger challenge and made a lot of difference, is thinking about all the consequences that this has. Like, for example, more than once we have already discussed whether mandatory PR review by another developer still makes sense. Nowadays we open a PR and it cannot be merged until someone different from the person who opened it goes there and gives approved, right? Now we have been asking ourselves whether this still makes sense, because the code was generated by someone and this someone is AI. The one who generated this code is the reviewer. So, we already have a review from someone who was not the one who did it. So, there are people who are very adept at removing mandatory code review by another developer. However, I already get kind of undecided because, okay, you did not type it. But you were the one who told it to do it that way. I do not think you are another person, you know? I do not think your review is as unbiased as people understand it is just because the AI generated the code and you reviewed it and they are two different entities. No, it did what you told it to do. So, even so, an external view makes sense, you know? This is a discussion that we have already had in a thread, already had in a call and it has not been decided yet. Even the people who are very adept, who say we have to remove the GitHub restriction already, that we have to put it to merge yes, without having to look. And there are me and other people who are more like, but let's discuss. Let's, wait a minute. And then there are other things like the development process itself. Now we no longer have in progress. What is in progress? In progress before was when I am coding. Usually, to enter in progress, the task was already well described and everything clear, it does not need to be done, ideally, at least. Nowadays, in progress is no longer generating the code, because AI generates the code in a few minutes. So, like, in progress ended up becoming the specification or the previous technical refinement, which is the moment of generating the PRD. So it gives the impression that now in progress is no longer generating the code, now it is generating the markdown, because generating the code is very quick. And then even thinking about process, in a Kanban board of life, the in progress column seems to be changing, the code review column that came after it, some people want to remove. So this has been more challenging, thinking about everything else. Okay, generating code is OK, we already understood that we need to, we already understood that it does it. Now, the change that this generates in a chain is what has been more difficult for us as a company, you know? All the teams are trying to do it in some way. Then we have some exchange moments to say: "Look, I did it this way, I did the whole specification of a task with PRD, TechSpec and test." Then another person comes and says: "No, wow, it is too annoying, I have to generate three documents." Sometimes it is a simple change, just vibe code it at the root and it will manage, because the code is good. So, I think the change, beyond code, has been the biggest challenge and I see that for several people, because it generates the code indeed. And all right, it is being OK.

And speaking of this Markdown, that you mentioned you generate, how many artifacts do you basically generate like this, when you are going to do or solve a task.

Regarding these markdown documents, right? The PR, right? Ideally, as I said, there is the whole process and there are times when we think we do not need all of them. But the complete one would be the generation of the first document that we call PRD, right? Which is the product requirements document. And then this is where we would have the definition of what needs to be done. So there are things there like ah, I need a screen that shows the information, then it puts some user stories, puts there a, all in text, right? A user journey, where he clicks here, opens such thing and everything else. Functional requirement, non-functional requirement, limitations, something much more like it were the end of a requirements elicitation itself, in the product definition there and it is expected that this document, actually, is generated by our product managers. But sometimes this has varied a little from team to team. There are times when my PM, for example, generates it. There are other times when I am like: "Ah, let me generate it already because I already understood, I will already generate the rest since these things were already

said." But it starts with this anyway. After it, from the business definitions, we go to the definitions of how what that PRD is defining needs to be done will be done. So there, this second document, which is according to the spec kit or tech spec, is where we already go down the level a little. It is what I told you, if I am going to have to make the change in four repositories, because we have microservices, micro frontend, BFF, I don't know what, I think it is nice to make a tech spec for each one. Because they become, in a way, specialized and more specific in the code in the codebase they will have to change, but they talk to each other. So, I think it is much better than making, for example, a PRD and one tech spec, given that I will have to change four things. This tech spec, it has two paths, if it is just one, it will become huge, or it will become little detailed so it does not become huge. So, this second moment comes, which is a more technical document that already has code. It basically already comes written like this: I need to create a new route, the name of the route will be X, I will create a file in folder such with this content here. So, this is where it is almost as if we were already reviewing the PR, only in this new format. Instead of seeing the code already there generated, I already see a document saying what will be generated, but it is not only saying it in natural language, actually showing the code that it will, in quotation marks, copy and paste there to that file that it is specifying. That is why the tech spec goes down much more in level and that is why then we already understand that unanimously it should be the responsibility of a developer to generate it, precisely because it will be the person who will understand. Even though, if we learn to generate a good tech spec, like, the PMs themselves will be able to generate it, because they will not need to review so well. Then comes what I told you, about evolving this prompt, evolving these instructions to generate all of this. So, I have this second one, which is the tech spec. Uh, the full flow, even the spec kit says that, after the tech spec, we generate a document or a little list of tasks, right, which is like, okay, I need to generate a route. But think in a very simple way, that for me to generate a route, I generate the unit test, right? To generate the unit test, leaving TDD aside, we first generate the code and then generate the tests to see if it is passing. So, you understand that there has to be a certain order in what will be generated and in what, a certain order in what will be generated for it to make sense. I don't know, you will have to generate two components. First you generate one and then you generate the other, because when you click on one, the other opens, and so on. So, the tasks already put it like this, look, of all those tasks that I said I would do in the tech spec, here is the order, here are three that I can do at the same time in parallel, after I finish these three I do this one. It is almost as if it assembled a tag, which is a sequence of ordered steps, right? That have dependency there for you to move to the next one, you have to finish the previous ones. And then for each line of this, which is just a listing, it still generates more files, which are a specific task, then here, like, the first task there was to generate route. So, then there will be a file called task 1 with the definitions of how to generate route. So, still until the very end, it is a lot of MD, actually, it becomes four, right? 4 Markdowns, but what we really need to review, what we usually review strongly, are the first two. The others are more for Claude to better understand how the code will be generated. And usually in this last one, the two, I do not check much. Then I already trust that it will do according to what was justified and leave it to check better when reviewing the PR. Then I will again look at the general code and then see if it is right. Then, as I told you, it is a lot of documents, there are people who already think that for a few simple things, you only do the PRD, only for me to understand what I have to do, from the PRD I already send it to generate the code. Or I do the Tech Spec, from the Tech Spec I send it to generate the code. This is something that we are still learning, there is this ideal, but still, it is not how we do it every time and it is not how we want to do it either, we are still discovering.

What is the fourth file that you generate?

It is a specific task. There is the third one, which is tasks, in plural, which is just an enumerated list really. And then each item of this list becomes a prompt file, then in this case, in the end, it is always a lot of files. It always generates several so that it understands what to do.

And what difficulties do you find when using AI?

As I said, to use itself I think it is easy, you just talk and it does it. So, interacting with a prompt there, absolutely I think anyone can do it, so much so that chat GPT exploded because of that, right? Anyone sends there what they want in a text message, it does it or it answers. So, depending on what you mean by use, use I think is easy. Now, as I had said, understanding how to integrate the use of AI in an optimized way, because we have already seen that only using it to generate code is not enough. It is not enough to use it to generate code and think that now you will have a, you will be 10 times faster than before. Because it is like in theory about traffic, when you solve a bottleneck and put everyone to pass quickly at the traffic light, there is probably a traffic light ahead that will get everyone stuck. Because it has to be something that is along this whole path. So in the same way, it is not enough just to generate code and then now everyone has to review this generated code and everyone who will review takes time, it did not help much that the code was generated quickly, if right afterward comes a step that blocks. So, I think the hardest part, as I had said, is understanding everything else. It is understanding how we actually make everything become optimized, because, from what I am understanding, just generating the code does not make it very fast. It makes the tasks go faster to a column on the side, but like, the final delivery, as I told you, sometimes the code is generated quickly, but it is generated quickly and wrong. And then, do I not take longer having to keep fixing it than if I had generated it right the first time? So there are several discussions that make me think like this, that the important thing is everything else, the whole development process itself, and not only the moment of generating PR.

And are there situations in which AI does not help?

That it does not help, then it is always a discussion. Like, where I work now we are always falling into the thought that if you asked AI to do something and it did not do it right, it is probably your fault. The first person we learned to blame is ourselves, because we did not give context, because we did not explain what it was supposed to do, because we did not say it right, because the prompt was not that good, I don't know what. So, normally we blame ourselves, because AI is the one who does it right and AI is the one who does everything right. But there are some things that it has not worked so well for us, such as generating a screen, generating screen code, it does not generate it so well, because then a different color comes, a border that it did not understand was another way. There are some things that we can do, again, it is our fault, because ah, it generated the button with the wrong border, but why didn't you say in the specification that this was the right border. And then these discussions come, you know, that if the AI is doing it wrong, it is your fault. And then, if you are not generating many more things with AI, it is because you are using it wrong. Something is wrong. Everyone said it is magic. So, I think it can do practically everything. Except not some things, then it really needs a lot of accompaniment, such as, for example, sometimes, I will do some refinement more at the very beginning, general PRD and such. And it does not understand all the business rules or it does not make some decisions that we make to define how the journey will be, ah, there is a super complex journey and a simpler one. Let's choose the simpler one now, because I think it will already make sense. So, like, it is very good for everything, but it also does not do anything by itself. Because we always have to be there beside it, so much so that this whole process of markdown, document and everything, everything, everything, is focused on generating things so that we can keep up with and validate what it is doing. Because it seems to be very intelligent, but it is very dumb, actually, because if you do not do it, if you do not say it right and do not give the right instruction, it understands wrong and does it wrong. So it is even a difficult answer, because it can do everything, but at the same time, by itself it does nothing.

Based on all this experience of yours, all this lived experience, what types of practices, guidelines or tools could improve the use of AI?

Uh, I would not even think much about tools yet, because I think nowadays there are already a lot of proper tools to create agents and the agent does everything. So like, tool for tool, we are already even very well served, because you can make a prompt for an agent that not only answers you,

but does other things too, right? But what I feel is missing today, and in conversation with other people that make me feel this is missing, is understanding how we use all of this. Because like, we went from very little to the extreme of options and ways of using AI and it is still not very clear the best way to use it, as I told you. It is not just talking about how to generate the code in the best way. Okay, the code was generated quickly, but there are hundreds of other bottlenecks that happen in the process too, how do we deal with them, you know? So, I would not say method so much, maybe practices. Uh, someone who already knew how to tell me like this: "Look, I have really tested it every way. And this whole spec kit only gives a lot of work. If you remove one of these documents, you already get a result that is, less or equal, at least, or something like that" something that makes sense for us to assume the risk. So, nowadays I miss these instructions on how I use all of this, because the tools are there. I think the problem is that there are too many tools. I am the one who is not knowing how to use them and it still generates this impostor syndrome that I cannot be completely productive the way absolutely everyone on YouTube says they are being, generates an entire website in two hours and I cannot. Why?